Ansible mit AWX im Unternehmen skalieren

Ein Erfahrungsbericht

GISA®
That's IT.

ls -tr

1. README.md
2. cmdb_inventory.py
3. bootstrap.yml
4. deploy-awx.yml
5. site.yml
6. BUGS.md
7. TODO.txt

README.md

- GISA GmbH
  - IT-Fullservice-Dienstleister aus Halle (Saale)
  - Energiewirtschaft, Hochschulen & Forschungseinrichtungen, öffentliche Auftraggeber und Industrie
  - Standorte in Leipzig, Berlin, Chemnitz, Cottbus
  - BSI-zertifizierter Rechenzentrumsverbund

Betriebssysteme

OS	Anzahl
Windows	~1500
SLES	~900
Ubuntu	~600
RHEL	~100
Oracle Linux	~100
AIX / Solaris	~100

Plattformen

Name	Anzahl
VMware	~2500
Ganeti	~500
OLVM	~100
HW	~500

cmdb_inventory.py

- zentrales CMDB-Inventar = single source of truth
- dynamisches Inventar mit Datenbankzugriff
- Fallback auf Inventardump im Fehlerfall
- Gruppen für Abweichungen/Erweiterungen

Hostinformationen

```
"srvr30210": {
      "ansible_host": "FQDN/IP",
      "vcpu": 4,
      "ram": 16,
      "disk": 200,
      "os": "deb13",
      "vlan": 1234,
      "roles": ["k8s_controlplane"]
 }
```

Gruppe

```
"role_k8s_controlplane": {
    "hosts": [
        "srvr30210",
        "srvr10244"
    ]
}
```

bootstrap.yml

- Grundkonfiguration von Servern ab Ende 2013
- Standardkonfiguration + ggf. Funktionsspezifika
- Erstellung neuer VMs aus Image
- Adaption auf HW-Server mittels Rescue ISO

deploy-awx.yml

- AWX-Instanz für das eigene Team
- Ziele
  - Konfigurationsdrift periodisch korrigieren
  - Ausführung von Playbooks delegieren

- docker-compose Installation + isolated node
- AWX-Image mit eigenen Erweiterungen
- Migration auf Kubernetes mit eigenem EE-Image

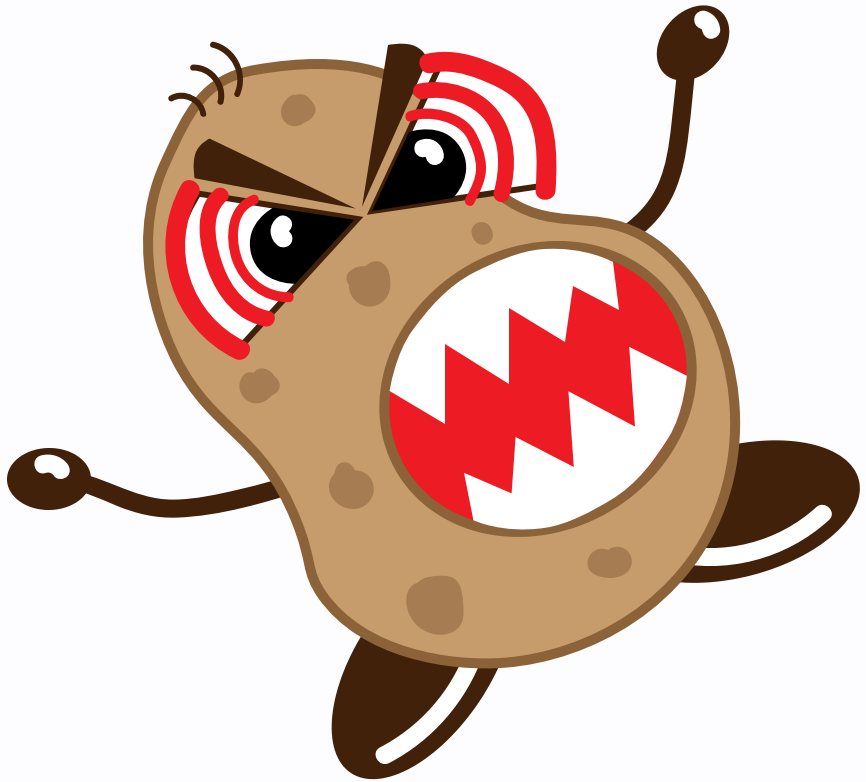
site.yml

- AWX Zugriff organisationsübergreifend geöffnet
- Ansible Playbooks enstehen in Eigenregie
- wenig Austausch zwischen Organisationen
- AWX als Abstraktionsebene für GUI-Freunde

Organisationen	8
Templates	~700
Jobs pro Monat	~17000
Ausführungszeit pro Monat	~860 h

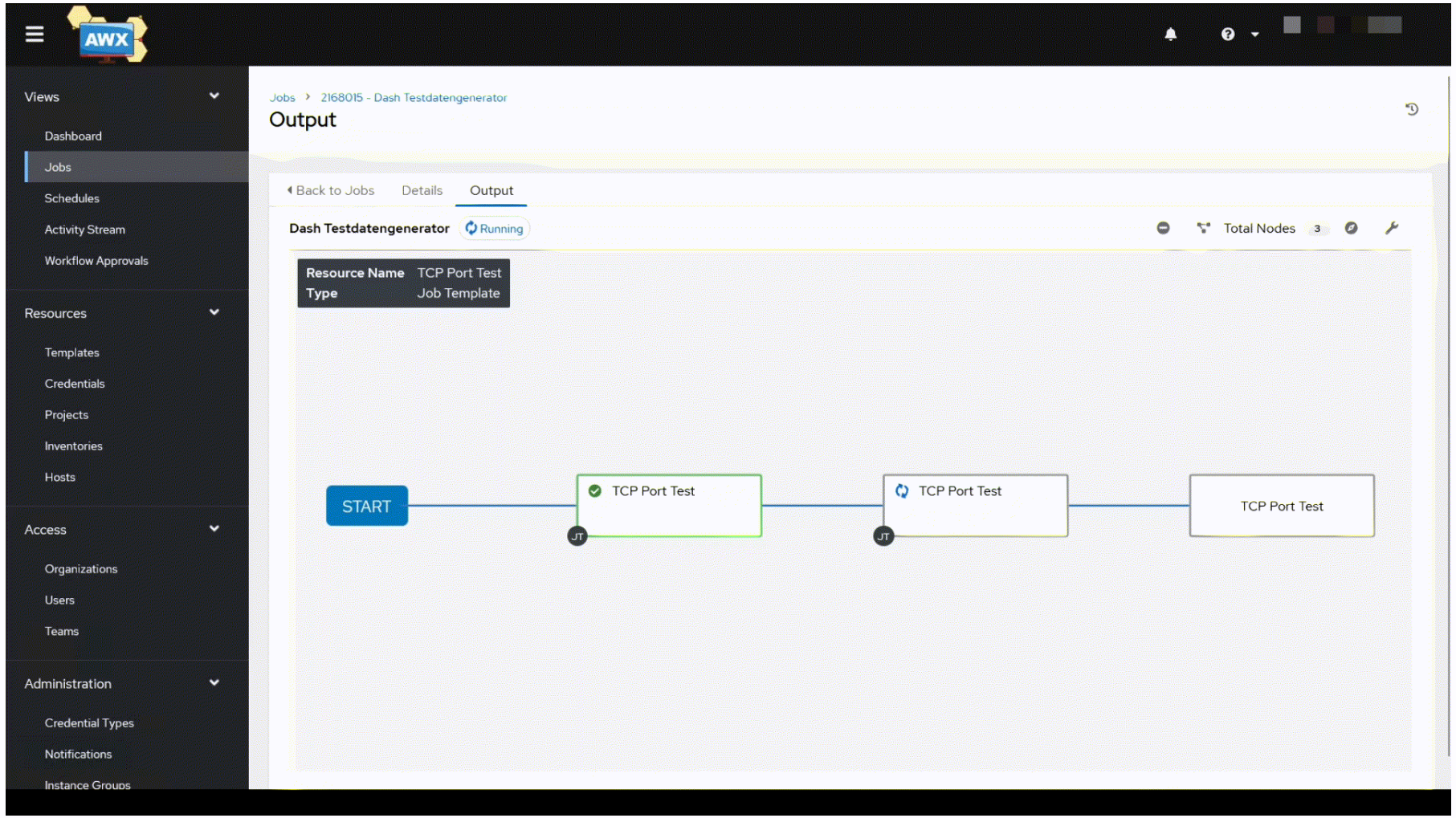
site.yml

BUGS.md



Jobfehler

- unkomfortable Fehlersuche im Frontend
- lange Inventarupdates bei vielen Jobdaten
- Eigenentwicklung eines Analysetools


AWX
Views
Dashboard
Jobs
Schedules
Activity Stream
Workflow Approvals
Resources
Templates
Credentials
Projects
Inventories
Hosts
Access
Organizations
Users
Teams
Administration
Credential Types
Notifications
Instance Groups
Jobs > 2168015 - Dash Testdatengenerator
Output
Back to Jobs
Details
Output
Dash Testdatengenerator
Running
Total Nodes 3
Resource Name TCP Port Test
Type Job Template
START
TCP Port Test
TCP Port Test
TCP Port Test
JT
JT

AWX Dash Templates Workflows Jobs Hosts

live

Workflow 2168019 - Dash Testdatengenerator

Job 2168020 - TCP Port Test: finished → Job 2168021 - TCP Port Test: finished → Job 2168022 - TCP Port Test: finished

Ubersicht Recaps Callbacks

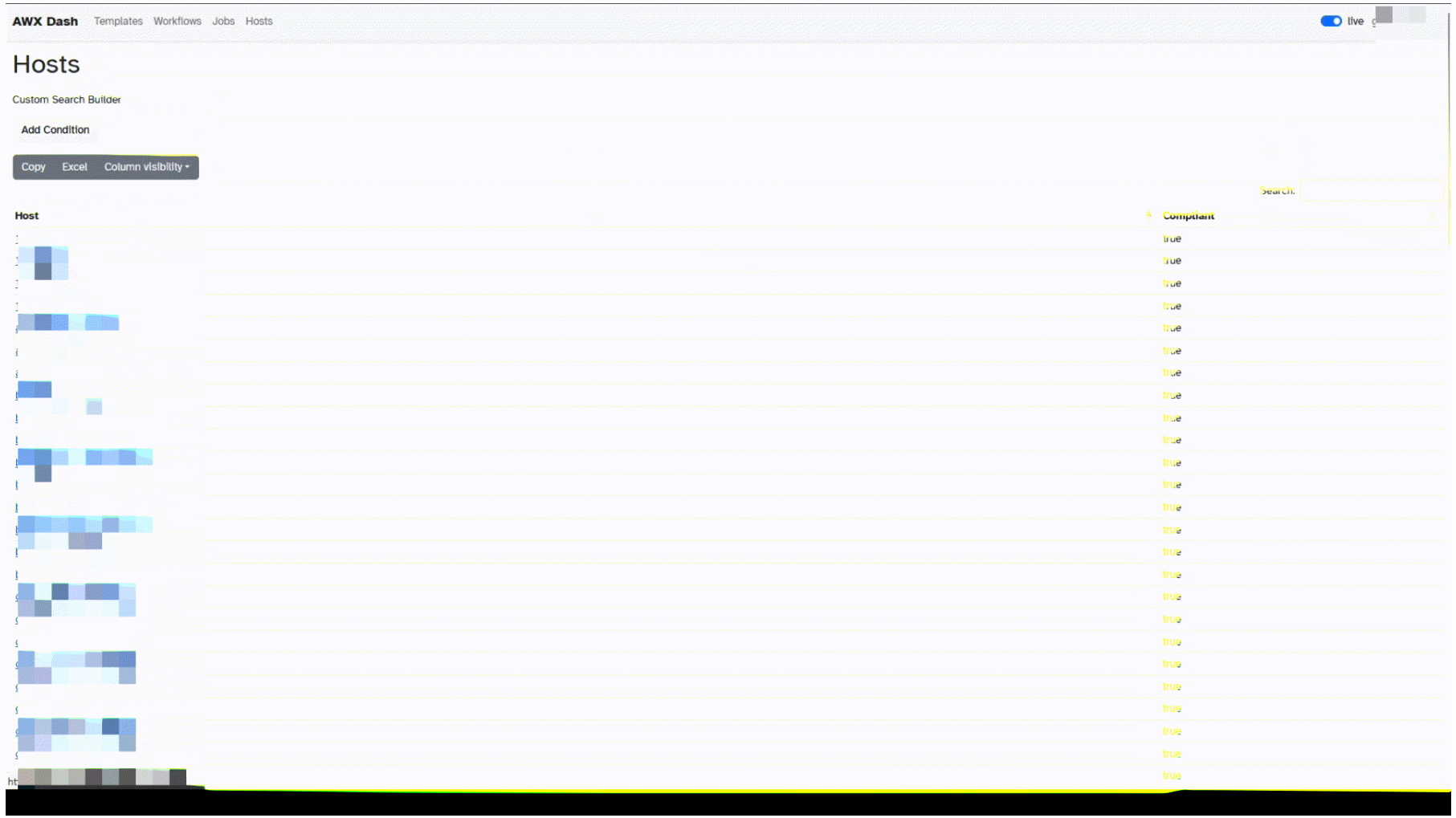
Custom Search Builder

Add Condition

Copy Excel Column visibility

Search:

ID	Job ID	Job Name	Task	Host	Status	Zeit
457565270	2168022	TCP Port Test	set_fact	localhost	failed	21.3.2025 13:44:55
457565269	2168022	TCP Port Test	set_fact	localhost	started	21.3.2025 13:44:54
457565268	2168022	TCP Port Test	set_fact	localhost	ok	21.3.2025 13:44:54
457565267	2168022	TCP Port Test	set_fact	localhost	started	21.3.2025 13:44:54
457565266	2168022	TCP Port Test	Prüfe ob Destination als Host im Inventar vorhanden ist	localhost	ok	21.3.2025 13:44:54
457565265	2168022	TCP Port Test	Prüfe ob Destination als Host im Inventar vorhanden ist	localhost	started	21.3.2025 13:44:54
457565264	2168022	TCP Port Test	Prüfe Destination auf Regex	localhost	skipped	21.3.2025 13:44:54
457565263	2168022	TCP Port Test	Prüfe Destination auf Regex	localhost	started	21.3.2025 13:44:54
457565262	2168022	TCP Port Test	Gathering Facts	localhost	ok	21.3.2025 13:44:54
457565261	2168022	TCP Port Test	Gathering Facts	localhost	started	21.3.2025 13:44:52
457565260	2168022	TCP Port Test	Prüfe, dass nur ein Zielserver angegeben ist	localhost	ok	21.3.2025 13:44:52
457565259	2168022	TCP Port Test	Prüfe, dass nur ein Zielserver angegeben ist	localhost	started	21.3.2025 13:44:52
457565254	2168021	TCP Port Test	Sende mail an {{ mail }}	localhost	started	21.3.2025 13:44:38
457565253	2168021	TCP Port Test	Prüfe Email	localhost	skipped	21.3.2025 13:44:38
457565252	2168021	TCP Port Test	Prüfe Email	localhost	started	21.3.2025 13:44:38
457565251	2168021	TCP Port Test	Setze EMail	localhost	skipped	21.3.2025 13:44:38
457565250	2168021	TCP Port Test	Setze EMail	localhost	started	21.3.2025 13:44:38
457565248	2168021	TCP Port Test	Extrahiere Mail aus g-Account	localhost	skipped	21.3.2025 13:44:38
457565247	2168021	TCP Port Test	Extrahiere Mail aus g-Account	localhost	started	21.3.2025 13:44:38
457565246	2168021	TCP Port Test	Lade Zugangsdaten AWX Mail	localhost	skipped	21.3.2025 13:44:38
457565249	2168021	TCP Port Test	Lade Zugangsdaten AWX Mail	localhost	started	21.3.2025 13:44:38
457565242	2168021	TCP Port Test	set_fact	localhost	skipped	21.3.2025 13:44:38


AWX Dash
Templates
Workflows
Jobs
Hosts
Hosts
Custom Search Builder
Add Condition
Copy
Excel
Column visibility
Search:
Host
Compliant
true

AWX Dash Templates Workflows Jobs Hosts

live

Template 648 - TCP Port Test

Jobs Hosts Infos

Custom Search Builder

Add Condition

Copy Excel Column visibility

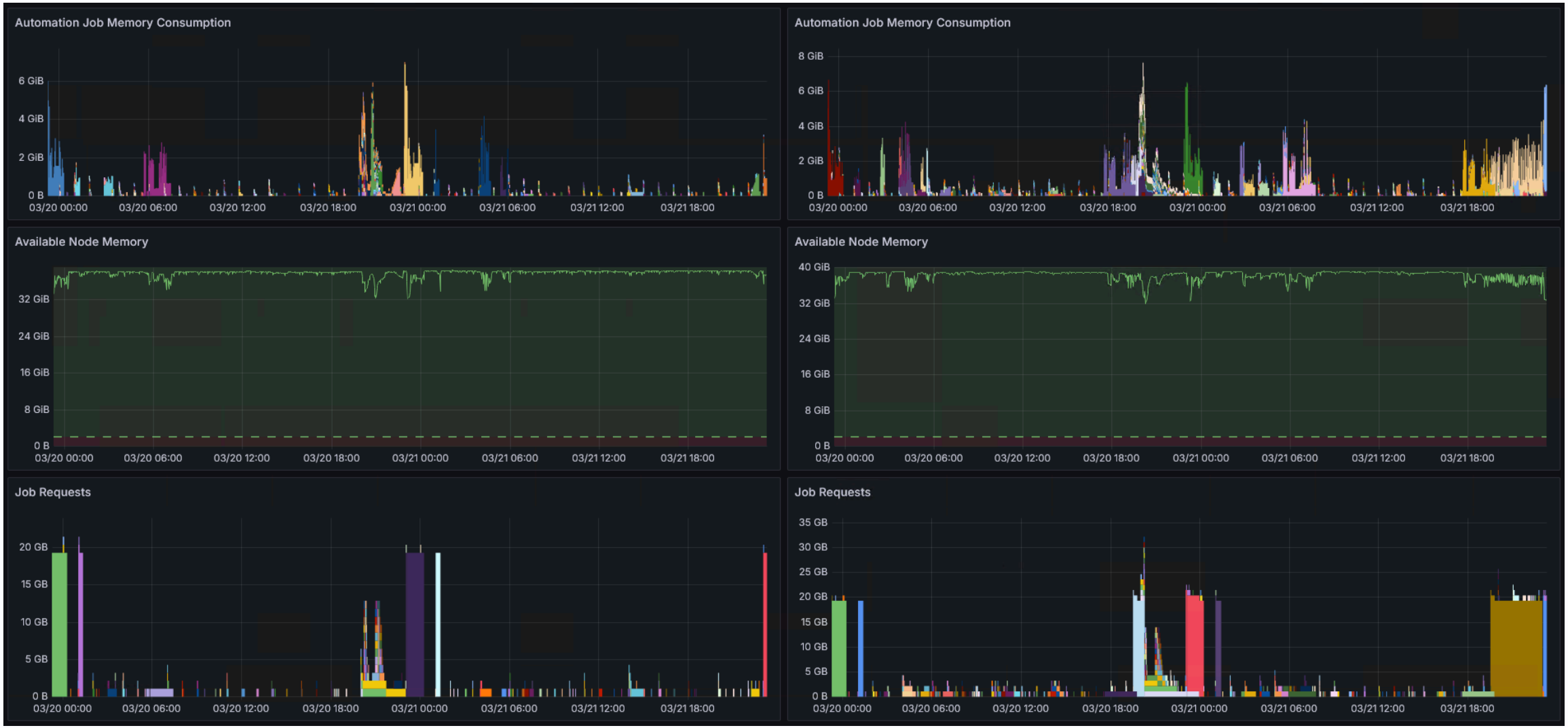
Search:

Job ID	Tags	Sicherheitsrelevant	Benutzer	Schedule	Workflow	Startzeit	Endzeit	Status	Limit
2168022	all	false	g436327_a		Dash Testdatengenerator	21.3.2025 13:44:52	21.3.2025 13:44:55	finished	
2168021	all	false	g436327_a		Dash Testdatengenerator	21.3.2025 13:44:34	21.3.2025 13:44:39	finished	
2168020	all	false	g436327_a		Dash Testdatengenerator	21.3.2025 13:44:17	21.3.2025 13:44:21	finished	
2168018	all	false	g436327_a		Dash Testdatengenerator	21.3.2025 13:43:33	21.3.2025 13:43:35	finished	
2168017	all	false	g436327_a		Dash Testdatengenerator	21.3.2025 13:43:16	21.3.2025 13:43:20	finished	
2168016	all	false	g436327_a		Dash Testdatengenerator	21.3.2025 13:42:59	21.3.2025 13:43:03	finished	
2168011	all	false	g436327_a		Dash Testdatengenerator	21.3.2025 13:37:44	21.3.2025 13:37:47	finished	
2168010	all	false	g436327_a		Dash Testdatengenerator	21.3.2025 13:37:27	21.3.2025 13:37:31	finished	
2168007	all	false	g436327_a		Dash Testdatengenerator	21.3.2025 13:37:10	21.3.2025 13:37:14	finished	
2162948	all	false	g436327_a		Dash Testdatengenerator	19.3.2025 14:55:05	19.3.2025 14:55:07	finished	
2162947	all	false	g436327_a		Dash Testdatengenerator	19.3.2025 14:54:49	19.3.2025 14:54:53	finished	
2162946	all	false	g436327_a		Dash Testdatengenerator	19.3.2025 14:54:33	19.3.2025 14:54:37	finished	
2162944	all	false	g436327_a			19.3.2025 14:54:06	19.3.2025 14:54:08	finished	
2162936	all	false	g436327_a		Dash Testdatengenerator	19.3.2025 14:48:17	19.3.2025 14:48:20	finished	
2162935	all	false	g436327_a		Dash Testdatengenerator	19.3.2025 14:48:01	19.3.2025 14:48:05	finished	
2162934	all	false	g436327_a		Dash Testdatengenerator	19.3.2025 14:47:45	19.3.2025 14:47:49	finished	
2162920	all	false	g436327_a		Dash Testdatengenerator	19.3.2025 14:34:48	19.3.2025 14:34:51	finished	
2162919	all	false	g436327_a		Dash Testdatengenerator	19.3.2025 14:34:32	19.3.2025 14:34:36	finished	
2162918	all	false	g436327_a		Dash Testdatengenerator	19.3.2025 14:34:15	19.3.2025 14:34:19	finished	
2162916	all	false	g436327_a		Dash Testdatengenerator	19.3.2025 14:31:07	19.3.2025 14:31:11	finished	
2162897	all	false	g436327_a		Dash Testdatengenerator	19.3.2025 14:30:50	19.3.2025 14:30:54	finished	
2056155	all	false	g438482_m			5.2.2025 13:38:18	5.2.2025 13:38:43	finished	
1987103	all	false	g527194_m			9.1.2025 12:03:57	9.1.2025 12:04:01	finished	

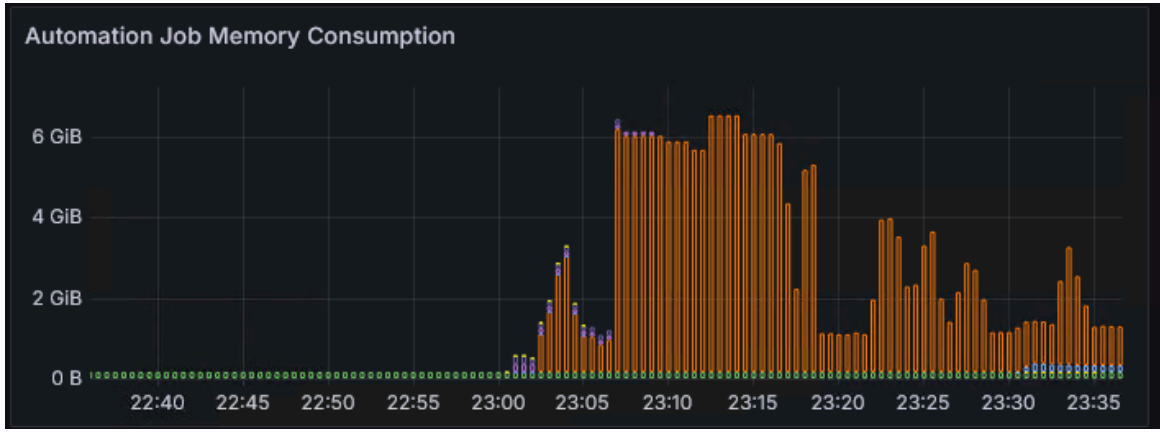
Ressourcenbedarf

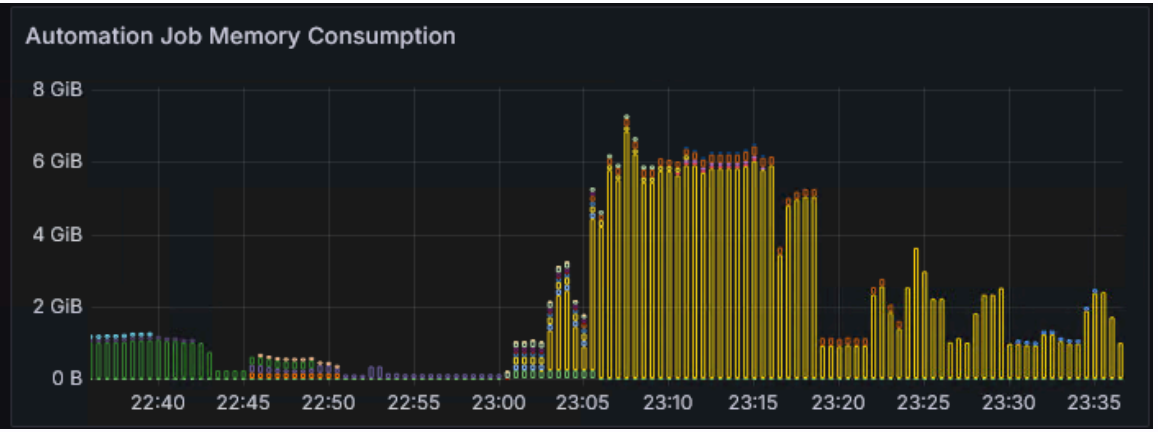
AWX Jobs
25
20
15
10
5
0
03/20 00:00
03/20 03:00
03/20 06:00
03/20 09:00
03/20 12:00
03/20 15:00
03/20 18:00
03/20 21:00
03/21 00:00
03/21 03:00
03/21 06:00
03/21 09:00
03/21 12:00
03/21 15:00
03/21 18:00
03/21 21:00
Job Requests
50 GB
40 GB
30 GB
20 GB
10 GB
0 B
Automation Memory
12 GiB
8 GiB
4 GiB
0 B

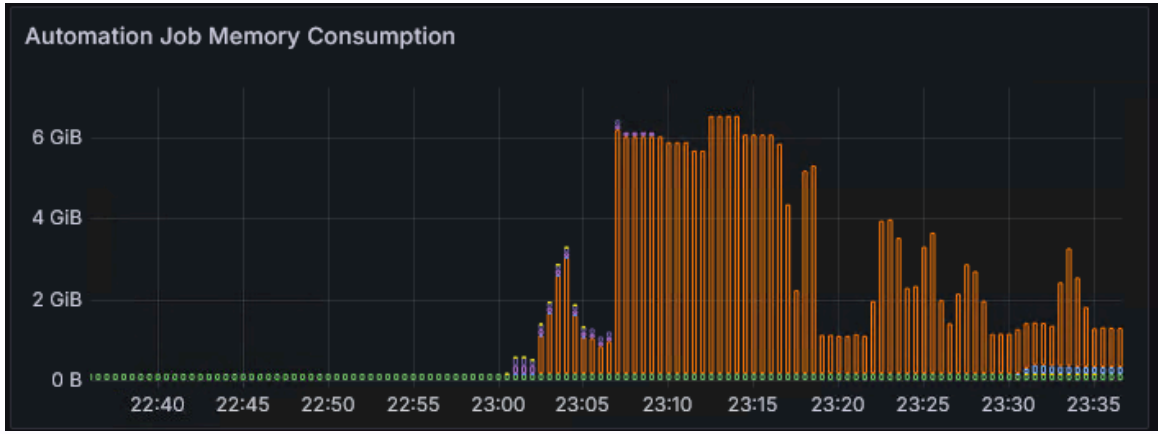
Load 1/5/15
1.01
1.40
1.54
Load 1/5/15
4.39
46.5
55.5
Load 1
200
150
100
50
0
03/20 00:00
03/20 06:00
03/20 12:00
03/20 18:00
03/21 00:00
03/21 06:00
03/21 12:00
03/21 18:00
Load 1
175
150
125
100
75
50
25
0
03/20 00:00
03/20 06:00
03/20 12:00
03/20 18:00
03/21 00:00
03/21 06:00
03/21 12:00
03/21 18:00

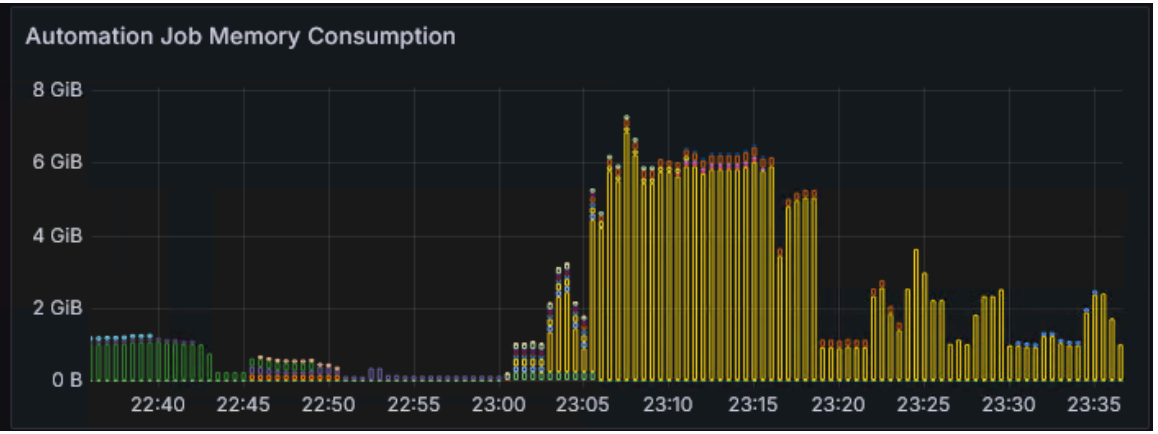

Automation Job Memory Consumption
6 GiB
4 GiB
2 GiB
0 B
03/20 00:00
03/20 06:00
03/20 12:00
03/20 18:00
03/21 00:00
03/21 06:00
03/21 12:00
03/21 18:00
Automation Job Memory Consumption
8 GiB
6 GiB
4 GiB
2 GiB
0 B
Available Node Memory
32 GiB
24 GiB
16 GiB
8 GiB
0 B
Available Node Memory
40 GiB
32 GiB
24 GiB
16 GiB
8 GiB
0 B
Job Requests
20 GB
15 GB
10 GB
5 GB
0 B
Job Requests
35 GB
30 GB
25 GB
20 GB
15 GB
10 GB
5 GB
0 B

- Teils massiver RAM Verbrauch bei vielen Hosts










```yaml
- name: "Prüfe ob group_vars existieren"
  stat:
    path: "{{ playbook_dir }}/group_vars/{{ item }}"
  with_items:
    - "{{ ansible_system }}"
    - "{{ ansible_system }}_{{ ansible_kernel }}"
    - "{% if cmdb_dnsdomainname == 'none' %}DNS{% else %}NAMESERVER{% endif %}_{{ ansible_default_ipv4.network|default(ansible_eth0.ipv4.network) }}"
    - "SYSLOG_{{ ansible_default_ipv4.network|default(ansible_eth0.ipv4.network) }}"
    - "MAIL_{{ ansible_default_ipv4.network|default(ansible_eth0.ipv4.network) }}"
    - "NTP_{{ ansible_default_ipv4.network|default(ansible_eth0.ipv4.network) }}"
    - "{{ ansible_distribution }}"
    - "{{ ansible_distribution }}_{{ ansible_distribution_major_version | default('undef') }}"
    - "{{ ansible_distribution }}_{{ ansible_distribution_major_version | default('undef') }}_{{ ansible_distribution_release }}"
    - "NIM_Master_{{ ansible_default_ipv4.network|default(ansible_eth0.ipv4.network) }}"
    - "STAGE_{{ cmdb.customer_name | regex_replace(' ', '_') }}"
  register: group_vars_stats
  delegate_to: "localhost"
  become: false
```

- AWX Scheduling in Kubernetes nicht mehr funktional
- Kubernetes Scheduling ist im Frontend nicht nachvollziehbar
- oom kills gefährden Betrieb

- oom kills gefährden Betrieb

```
"ansible-playboo invoked oom-killer: gfp_mask=0x400dc0(GFP_KERNEL_ACCOUNT|__GFP_ZERO), order=1, oom_score_adj=574"
"Out of memory: Killed process 2875266 (awx-manage) total-vm:1121724kB, anon-rss:196872kB, file-rss:0kB, shmem-rss:12kB, UID:1000 pgtables:580kB oom_score_adj:995"
"Out of memory: Killed process 221622 (awx-manage) total-vm:895492kB, anon-rss:170684kB, file-rss:0kB, shmem-rss:12kB, UID:1000 pgtables:508kB oom_score_adj:995"
"Out of memory: Killed process 223409 (awx-manage) total-vm:886532kB, anon-rss:157368kB, file-rss:0kB, shmem-rss:12kB, UID:1000 pgtables:496kB oom_score_adj:995"
"Out of memory: Killed process 1298608 (awx-manage) total-vm:255808kB, anon-rss:148176kB, file-rss:0kB, shmem-rss:0kB, UID:1000 pgtables:408kB oom_score_adj:995"
```

- oom kills gefährden Betrieb

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```

```
1000     1291346  0.0  0.5 242696 137240 ?       Sl   Oct22   1:04  |       \_ /var/lib/awx/venv/awx/bin/python3.11 /usr/bin/awx-manage run_callback_receiver
1000     1298518  0.0  0.0      0     0 ?        Z    Oct22   3:22  |           \_ [awx-manage] <defunct>
1000     1298577  0.0  0.0      0     0 ?        Z    Oct22   3:23  |           \_ [awx-manage] <defunct>
1000     1298578  0.0  0.0      0     0 ?        Z    Oct22   3:24  |           \_ [awx-manage] <defunct>
1000     1298608  0.0  0.0      0     0 ?        Z    Oct22   3:23  |           \_ [awx-manage] <defunct>
```

TODO.txt

- AWX als Ausführungsengine hinter flexiblerem Frontend
- Lose Kopplung von AWX-Templates mit Event-Driven-Ansible

Fragen?